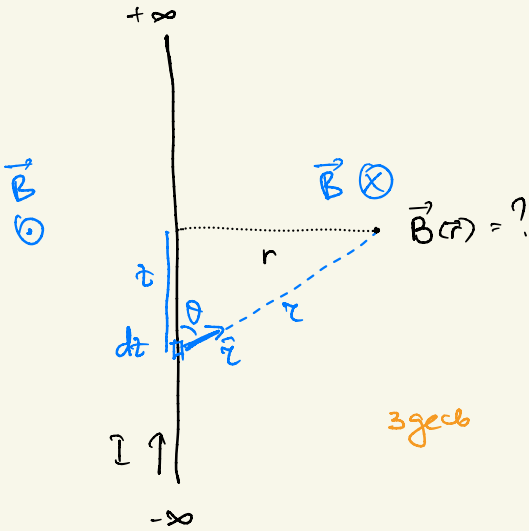


Ex1: Hallita $\vec{B}(r)$ gna tota ka dactonezno gauthom probe:



$$dB = \frac{\mu_0 I}{4\pi} \frac{dz \times \vec{r}}{r^2} \quad |d\vec{z}| |\vec{r}| \sin\theta$$

$$B = \int dB = \int \frac{\mu_0 I}{4\pi} \frac{dz \times \vec{r}}{r^2}$$

$$= \frac{\mu_0 I}{4\pi} \int \frac{dz (\sin\theta)}{(z^2 + r^2)}$$

3geob $\sin\theta = \frac{r}{z} = \frac{r}{\sqrt{r^2 + z^2}} = \frac{r}{(r^2 + z^2)^{1/2}}$

$$B = \frac{\mu_0 I}{4\pi} \int \frac{(dz) r}{(r^2 + z^2)^{3/2}}$$

$$= \frac{\mu_0 I r}{4\pi} \int_{-\infty}^{+\infty} \frac{dz}{(r^2 + z^2)^{3/2}}$$

$$= \frac{\mu_0 I r}{4\pi} \frac{z}{r^2 (z^2 + r^2)^{1/2}} \Big|_{-\infty}^{+\infty}$$

$$= \frac{\mu_0 I}{4\pi r} \frac{z}{\sqrt{z^2 + r^2}} \Big|_{-\infty}^{+\infty}$$

limit mathematics
(Calculus I)

$$\lim_{z \rightarrow a} \frac{z}{\sqrt{z^2 + r^2}} = \lim_{z \rightarrow a} \frac{1}{\sqrt{1 + \frac{r^2}{z^2}}} = \frac{1}{\sqrt{1 + 0}}$$

Now make $a = \infty$

$$= \frac{\mu_0 I}{4\pi r} \left(\frac{+a}{\sqrt{a^2 + r^2}} \right) - \left(\frac{-a}{\sqrt{a^2 + r^2}} \right)$$

